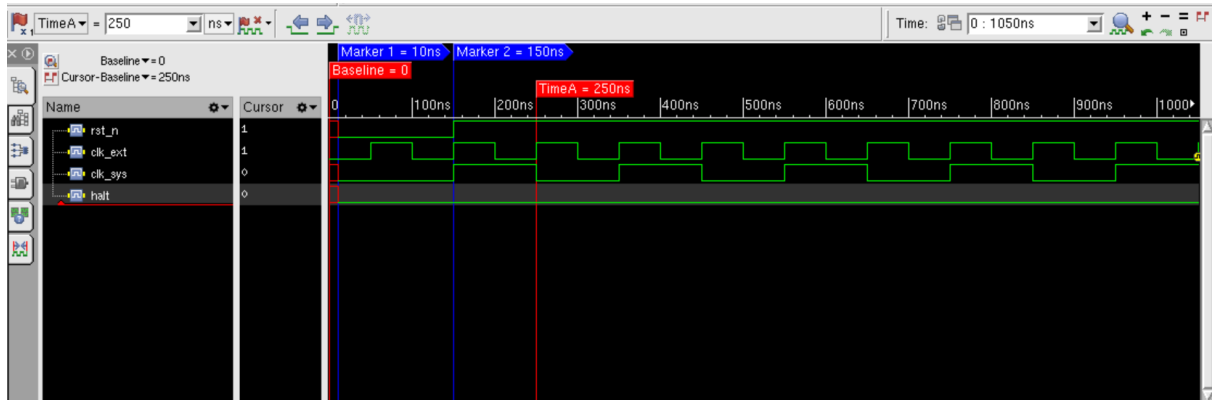


Lab 07: system clock => 시스템 클럭

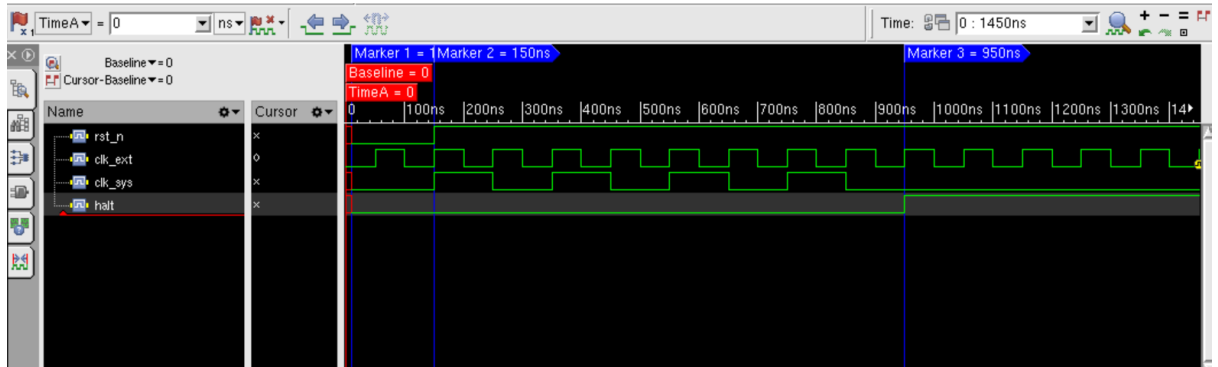
1. 주어진 설계파일, 테스트벤치파일 작성 후 시뮬레이션
2. 시뮬레이션하여 clk_sys가 clk_ext의 2분주인지 확인한다.



시뮬레이션을 통해 clk_sys가 clk_ext의 2분주임을 확인가능.

(8번의 외부 클럭 동안 시스템 클럭 관찰 : 4번의 클럭 상승 에지와 4번의 하강 에지)

3. comment #2를 추가하고 다시 시뮬레이션 한 결과



950ns부터 halt(정지신호)가 1이 됨.

(4번의 외부 클럭 동안 시스템 클럭 관찰 : 상승 에지와 하강 에지 모두 없음)

4. 완성품 복사 + git checkin

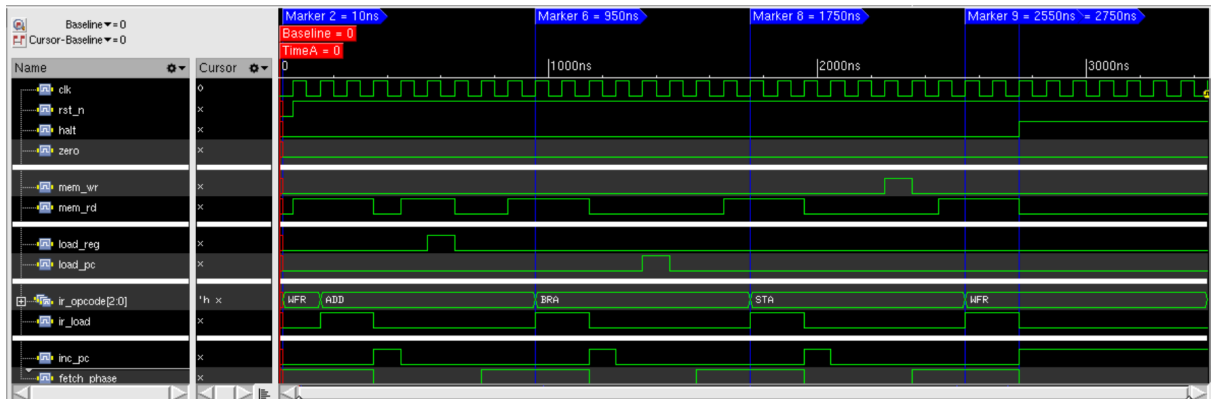
```
[std001@ky-cds-semi sim]$ cd ..
[std001@ky-cds-semi lab07-sysclk]$ cp sysclk_blank.sv ../lab00-design/sysclk.sv
[std001@ky-cds-semi lab07-sysclk]$ git status
On branch main
Your branch is ahead of 'origin/main' by 4 commits.
(use "git push" to publish your local commits)
```

```
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
    modified:   ../lab06-ir/cpu_pkg_blank.sv
    modified:   sysclk_blank.sv
    modified:   tb_sysclk_blank.sv
```

```
[std001@ky-cds-semi lab07-sysclk]$ git add sysclk_blank.sv
[std001@ky-cds-semi lab07-sysclk]$ git add ../lab00-design/sysclk.sv
[std001@ky-cds-semi lab07-sysclk]$ git commit -m "lab07: done"
[main 0cb012f] lab07: done
2 files changed, 38 insertions(+), 9 deletions(-)
create mode 100644 lab00-design/sysclk.sv
[std001@ky-cds-semi lab07-sysclk]$ git push
```

Lab 08: control FSM (제어)

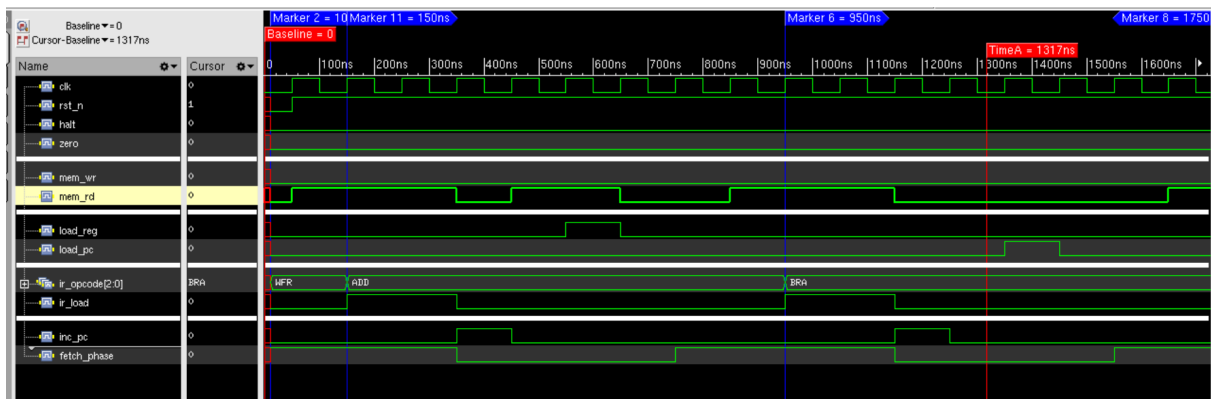
1. 설계파일, 테스트벤치파일 작성 후 시뮬레이션



각 opcode의 제어신호를 확인할 수 있다.

#10ns까지는 rst_n = 0; ir_opcode = WRF; zero = 0;

@(posedge clk); rst_n = 1이 됨. //상승에지 #50ns



2. 표와 비교하여 파형을 분석하기.

time	rst_n	ir_op code	mem_ rd	ir_ load	inc_ pc	load_ reg	load_ pc	mem_ wr	halt
10	1	WFR	1	0	0	0	0	0	0
150	1	ADD	1	1	0	0	0	0	0
350	1	ADD	0	0	1	0	0	0	0
950	1	BRA	1	1	0	0	0	0	0
1150	1	BRA	0	0	1	0	0	0	0
1750	1	STA	1	1	0	0	0	0	0
2250	1	WFR	1	0	0	0	0	0	0
2750	1	WFR	0	0	1	0	0	0	1

3. 완성품 복사 + git checkin

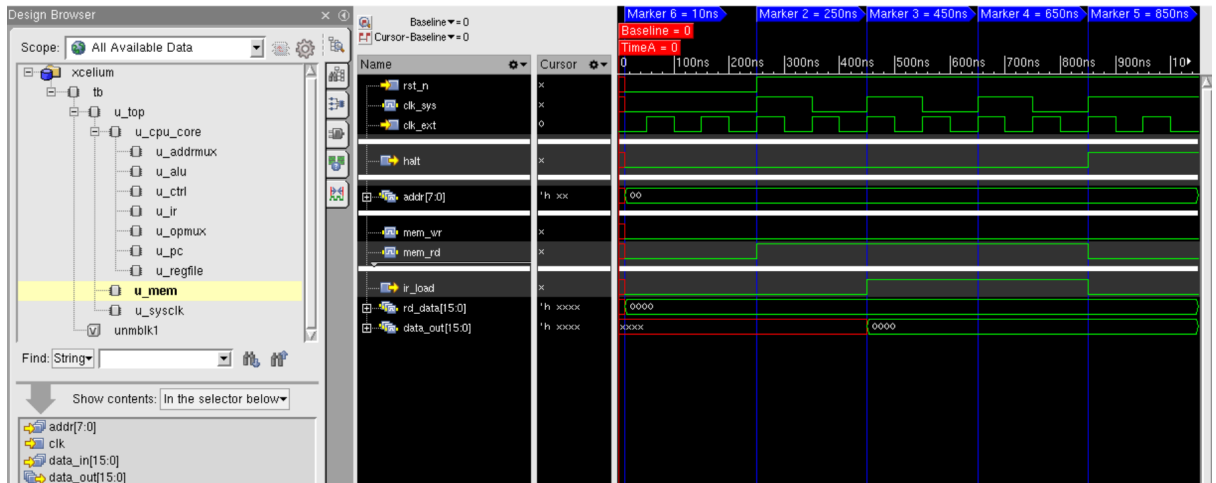
```
[std001@ky-cds-semi lab08-ctrl]$ cp cpu_pkg_blank.sv ../lab00-design/cpu_pkg.sv
[std001@ky-cds-semi lab08-ctrl]$ cp control_blank.sv ../lab
lab00-design/      lab03-mux/      lab06-ir/      lab09-core/      lab12-prog_movsum/
lab01-regfile/    lab04-pc/      lab07-sysclk/  lab10-prog_ctrl/
lab02-alu/        lab05-mem/     lab08-ctrl/    lab11-prog_arith/
[std001@ky-cds-semi lab08-ctrl]$ cp control_blank.sv ../lab00-design/control.sv
[std001@ky-cds-semi lab08-ctrl]$ git status
On branch main
Your branch is ahead of 'origin/main' by 5 commits.
(use "git push" to publish your local commits)

Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   ../lab06-ir/cpu_pkg_blank.sv
        modified:   ../lab07-sysclk/tb_sysclk_blank.sv
        modified:   control_blank.sv
        modified:   cpu_pkg_blank.sv
        modified:   tb_control_blank.sv
        modified:   ../lab09-core/tb_cpu_core_blank.sv

[std001@ky-cds-semi lab08-ctrl]$ git add cpu_pkg_blank.sv control_blank.sv
[std001@ky-cds-semi lab08-ctrl]$ git add ../lab00-design/cpu_pkg.sv
[std001@ky-cds-semi lab08-ctrl]$ git add ../lab00-design/control.sv
[std001@ky-cds-semi lab08-ctrl]$ git commit -m "lab08: done"
[main acfff77] lab08: done
4 files changed, 172 insertions(+), 71 deletions(-)
create mode 100644 lab00-design/control.sv
create mode 100644 lab00-design/cpu_pkg.sv
[std001@ky-cds-semi lab08-ctrl]$ git push
remote: Repository not found.
fatal: repository 'https://github.com/ksdcsemi-class/microcpu-kyu-2026-1-Dorociel.git/' not found
```

Lab 09: CPU Core + Top 조립

- 주어진 설계파일, 테스트벤치파일 작성 후 시뮬레이션



- 표와 비교하여 파형을 확인하기.

// time	rst_n	clk_sys	halt	state	addr	mem_rd	mem_wr	
// 50	0	0	0	INST_ADDR	00	0	0	
// 150	0	0	0	INST_ADDR	00	0	0	
// 250	1	1	0	INST_FETCH	00	1	0	#1
// 350	1	0	0	INST_FETCH	00	1	0	
// 450	1	1	0	INST_LOAD	00	1	0	
// 550	1	0	0	INST_LOAD	00	1	0	
// 650	1	1	0	IDLE	00	1	0	
// 750	1	0	0	IDLE	00	1	0	
// 850	1	1	1	OP_ADDR	00	0	0	

10ns후 rst_n=0; clk_ext 3번째 상승에지(rst_n= 1;)

halt = 1; 루프 종료.

- 완성품 복사 + git checkin

```
[std001@ky-cds-semi sim]$ cd ..
[std001@ky-cds-semi lab09-core]$ cp cpu_core_blank.sv ../lab00-design/cpu_core.sv
[std001@ky-cds-semi lab09-core]$ cp cpu_top_blank.sv ../lab00-design/cpu_top.sv
[std001@ky-cds-semi lab09-core]$ git status
On branch main
Your branch is ahead of 'origin/main' by 6 commits.
  (use "git push" to publish your local commits)

Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   ../lab06-ir/cpu_pkg_blank.sv
        modified:   ../lab07-sysclk/tb_sysclk_blank.sv
        modified:   ../lab08-ctrl/tb_control_blank.sv
        modified:   cpu_core_blank.sv
        modified:   cpu_top_blank.sv
        modified:   sim/lab09_blank.f
        modified:   tb_cpu_core_blank.sv
```

```
[std001@ky-cds-semi lab09-core]$ git add cpu_core_blank.sv cpu_top_blank.sv
[std001@ky-cds-semi lab09-core]$ git commit -m "lab09: done"
[main f3125af] lab09: done
 2 files changed, 37 insertions(+), 94 deletions(-)
[std001@ky-cds-semi lab09-core]$ git push
remote: Repository not found.
fatal: repository 'https://github.com/ksdcsemi-class/microcpu-kyu-2026-1-Dorociel.git/' not found
[std001@ky-cds-semi lab09-core]$
```